WORKED SOLUTION

STEP I 2004, Question 7

Pure • Generating Functions

STEP I 2004, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q7

The key idea

A linear recurrence and a power series are two descriptions of the same object. Multiplying the series by $1 - 7x + 10x^2$ lines up each coefficient a_n with a_{n-1} and a_{n-2}, so the recurrence makes almost every coefficient of the product vanish. What survives is a polynomial, so f is a rational function. Partial fractions and geometric series then give a_n directly. Part (ii) is the same machine, once we have found p and q from the given values.

Solution

Part (i)

For $|x| < \frac{1}{5}$ the series converges, so we may multiply by powers of x and add series term by term:

$$\begin{aligned} \mathrm{f}(x) &= a_0 + a_1 x + \sum_{n=2}^{\infty} a_n x^n, \\ -7x\,\mathrm{f}(x) &= -7a_0 x - \sum_{n=2}^{\infty} 7a_{n-1} x^n, \\ 10x^2 \mathrm{f}(x) &= \sum_{n=2}^{\infty} 10a_{n-2} x^n. \end{aligned}$$

(In the second and third lines we have relabelled the index so that every sum is over x^n with $n \geq 2$.) Adding,

$$\mathrm{f}(x) - 7x\,\mathrm{f}(x) + 10x^2 \mathrm{f}(x) = a_0 + (a_1 - 7a_0)x + \sum_{n=2}^{\infty} (a_n - 7a_{n-1} + 10a_{n-2}) x^n.$$

Every bracket in the sum is zero by the recurrence, and $a_0 = 2$, $a_1 - 7a_0 = 7 - 14 = -7$. So

$$(1 - 7x + 10x^2)\,\mathrm{f}(x) = 2 - 7x.$$

Now $1-7x+10x^2=(1-2x)(1-5x)$, which is non-zero for $|x|<\frac{1}{5}$, so we may divide:

$$\mathrm{f}(x)=\frac{2-7x}{(1-2x)(1-5x)}.$$

The numerator is the sum of the two factors, $(1-5x)+(1-2x)=2-7x$, so

$$\mathrm{f}(x)=\frac{(1-5x)+(1-2x)}{(1-2x)(1-5x)}=\frac{1}{1-2x}+\frac{1}{1-5x},$$

as required.

For $|x|<\frac{1}{5}$ we have $|2x|<1$ and $|5x|<1$, so both fractions are sums of convergent geometric series:

$$\mathrm{f}(x)=\sum_{n=0}^{\infty}(2x)^n+\sum_{n=0}^{\infty}(5x)^n=\sum_{n=0}^{\infty}(2^n+5^n)x^n.$$

Ask yourself Why can we now read off a_n?

We have two power series that agree for every x in the interval $|x|<\frac{1}{5}$. A power series is determined by its values near 0: its coefficients can be recovered as $a_n=\mathrm{f}^{(n)}(0)/n!$. So the coefficients must match, and

$$a_n=2^n+5^n \quad \text{for all } n\geq 0.$$

(As a quick check, $a_0=1+1=2$ and $a_1=2+5=7$.)

Part (ii)

First we need p and q. Putting $n=2$ and $n=3$ into the recurrence:

$$\begin{aligned} b_2=pb_1+qb_0: &\quad 40=10p+5q, \quad \text{so } 2p+q=8;\\ b_3=pb_2+qb_1: &\quad 100=40p+10q, \quad \text{so } 4p+q=10.\end{aligned}$$

Subtracting, $2p=2$, so $p=1$ and $q=6$. The recurrence is $b_n=b_{n-1}+6b_{n-2}$.

Exactly as in part (i), multiplying by $1-x-6x^2$ kills every coefficient from x^2 onwards:

$$(1-x-6x^2)\,\mathrm{g}(x)=b_0+(b_1-b_0)x=5+5x.$$

Since $1-x-6x^2=(1-3x)(1+2x)$, which is non-zero for $|x|<\frac{1}{3}$,

$$\mathrm{g}(x)=\frac{5+5x}{(1-3x)(1+2x)}=\frac{A}{1-3x}+\frac{B}{1+2x},$$

where $5+5x=A(1+2x)+B(1-3x)$. Putting $x=\frac{1}{3}$ gives $\frac{20}{3}=\frac{5}{3}A$, so $A=4$; putting $x=-\frac{1}{2}$ gives $\frac{5}{2}=\frac{5}{2}B$, so $B=1$. Hence

$$\mathrm{g}(x)=\frac{4}{1-3x}+\frac{1}{1+2x}.$$

For $|x| < \frac{1}{3}$ we have $|3x| < 1$ and $|2x| < 1$, so expanding both as geometric series,

$$g(x) = \sum_{n=0}^{\infty} (4 \cdot 3^n + (-2)^n) x^n,$$

and comparing coefficients as before,

$$b_n = 4 \cdot 3^n + (-2)^n.$$

Checks

The formulae reproduce the given values: $4 + 1 = 5$, $12 - 2 = 10$, $36 + 4 = 40$, $108 - 8 = 100$. Computing a_n and b_n from the recurrences up to $n = 39$ agrees exactly with $2^n + 5^n$ and $4 \cdot 3^n + (-2)^n$. Numerically, at $x = 0.1$ the partial sums of the series give 3.25 for f and 6.547619… for g, matching $\frac{1}{0.8} + \frac{1}{0.5} = 3.25$ and $\frac{4}{0.7} + \frac{1}{1.2} = \frac{275}{42}$.

Common slips

- Forgetting that the recurrence only holds from $n = 2$: the x^0 and x^1 terms of the product must be worked out separately, and they give the numerator.
- In part (ii), using the recurrence with p and q still unknown, or writing the numerator as $b_0 + b_1 x$ rather than $b_0 + (b_1 - pb_0)x$.
- Factorising $1 - x - 6x^2$ with the wrong signs, which leads to 3^n and 2^n instead of 3^n and $(-2)^n$.

Beyond STEP

The series f is called the *generating function* of the sequence (a_n), and the method works for any linear recurrence with constant coefficients: the denominator is always built from the recurrence, and its factors give the "characteristic roots" (here 2 and 5). Generating functions are a central tool in combinatorics and probability, where probability generating functions turn sums of random variables into products. The step "equal power series have equal coefficients" is justified properly in a first analysis course; see Abbott, *Understanding Analysis*, for power series and their uniqueness.

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